

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Centre of Excellence in Artificial Intelligence
2. **Subject Name:** Algorithm Design with Python Lab **L-T-P:** 0-0-3 **Credits:** 2
3. **Pre-requisites:** Programming and Data Structures, Programming and Data Structures Laboratory
4. **Syllabus and reference books:**

Syllabus:

- Introduction to programming in Python – defining functions in python using generators, decorators and lambda functions, recursion, analyzing memory and time complexity, debugging, exception handling and unit-testing,
- Fundamental data structures in Python – lists, dictionaries, sets and tuples.
- Python libraries for computation, data analysis and visualization – Numpy, Pandas, Matplotlib and Scipy.
- Classes and Objects – defining and initializing classes, instantiating classes through objects, inheritance, polymorphism, encapsulation and abstraction techniques using classes and objects.
- Sorting algorithms – quick sort, merge sort, insertion sort etc.
- String matching algorithms - Rabin-Karp algorithm, Knuth-Morris-Pratt algorithm, Boyer-Moore algorithm etc.
- Tree Algorithms – representing and traversing trees, binary, m-ary and AVL trees, binary search trees and B-trees, heaps and heap sort,
- Graph algorithms - representation and traversal of graphs, finding graph shortest paths and minimum spanning trees.

Reference Books:

1. Mark Lutz, *Learning Python*, O'Reilly Inc, (2013)
2. Allen B. Downey, *Think Python*, O'Reilly Inc. (2012).
3. Zed A. Shaw, *Learn Python 3 the Hard Way*, Addison Wesley (2017)
4. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, *Data Structures and Algorithms in Python*, John Wiley & Sons, (2013).
5. Bradley L. Miller and David L. Ranum, *Problem Solving with Algorithms and Data Structures using Python*, Franklin, Beedle & Associates, Inc. (2006)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Python preliminaries	3
2.	Data structures in Python	3
3.	Numerical computation with NumPy; data analysis with pandas.	3
4.	Scientific computing with SciPy, plotting and visualization with Matplotlib	3
5.	Classes and objects	3
6.	Sorting algorithms	3
7.	String matching algorithms	3
8.	Tree representation and traversal, binary, m-ary and AVL trees	3
9.	Binary search trees, B-trees, heaps and heap sort	3
10.	Graph representation and graph traversal algorithms	3
11.	Minimum spanning tree algorithms	3
12.	Graph shortest path algorithms	3
Total number of hours		36